

Streamlining Neural Certificate Synthesis

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July 17, 2026



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The verification problem

system $\models$ spec $\forall$ runs satisfy the property

- Continuous state $\mathcal{X} \subseteq \mathbb{R}^d$ — cannot enumerate
- Discrete time
- **Stochastic**: interested in expected behaviour
- Varying degrees of system access

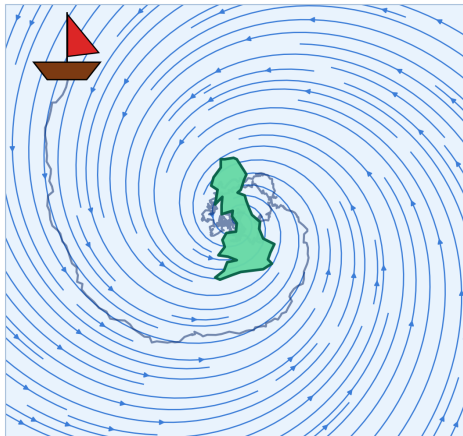
Certificates

A certificate $V : \mathcal{X} \rightarrow \mathbb{R}$ turns a $\forall$ -over-trajectories question into a local inequality.

Different specifications require different certificate types, and different certificate types require different conditions to hold.

- Reachability
- Safety
- Reach-Avoid

Reaching Great Britain on a stochastic tide



Does every path reach **home** in finite expected time?

Certificates

A certificate $V : \mathcal{X} \rightarrow \mathbb{R}$ turns a $\forall$ -over-trajectories question into a local inequality.

Reaching the “goal” set in finite expected time is called *termination*.

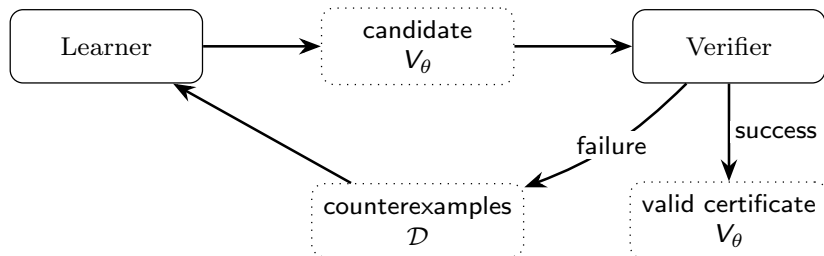
Ranking supermartingale condition

$$\mathfrak{D}(x) = \mathbb{E}[V(x') \mid x] - V(x) + \varepsilon \leq 0 \quad \forall x \in \mathcal{X}$$

How do we obtain a valid V ?

CEGIS: synthesise certificates from data

Learn V as a neural network; a sound verifier checks $\forall x$ and returns counterexamples to retrain.



Two goals inside one loop

Crucially, there are two distinct jobs:

- Find counterexamples to retrain on (**refutation**)
- Soundly verify all states (**verification**)

The bottleneck

Verification is expensive, yet the verifier does both jobs!

Hypothesis

A **cheap refuter** to find counterexamples will reduce reliance on the **sound verifier** and speed up the CEGIS pipeline.

Symbolic solvers

- Strong **white-box** assumption
- **Exact** encoding of the dynamics and certificate
- **Non-transcendental** functions only — no $\sin$, $\tanh$, ...
- Continuous noise: over-approximate the distribution with **bins**¹

Issues These methods scale poorly with number of variables/constraints to check: dimensionality, noise discretisation, etc. all increase this.

¹Lechner et al., *Stability Verification in Stochastic Control Systems*, AAAI 2022

Sampling

Minimal **black-box** assumption: a sample oracle plus a regularity (Lipschitz) bound.

Confidence bound

$$\text{CB} = \underbrace{\text{statistical error}}_{\downarrow \text{more samples}} + \underbrace{\text{discretisation error}}_{\downarrow \text{refine the grid}}$$

upper bound = empirical mean + CB

lower bound = empirical mean - CB

The discretisation error hinges on the Lipschitz constant

Lipschitz constant

$$\|g(x) - g(x')\| \leq L_g \|x - x'\|$$

$$\text{discretisation error} = L_{\mathcal{D}} \cdot \text{Diam}(B)$$

Two ways to shrink it:

- smaller cells $\rightarrow$ *grid refinement* (box count $N \approx (L_{\mathcal{D}}/\varepsilon)^d$)
- a **tighter** $L_{\mathcal{D}}$ $\rightarrow$ discretise less, everywhere

Results: a tighter $L_{\mathfrak{D}}$

d	L_f	spectral product	LipBaB	Our method
2	0.54	1.10	0.37	0.27
2	0.60	1.60	0.52	0.38
3	0.45	1.16	0.55	0.42
4	0.55	1.78	0.92	0.60
4	0.51	3.11	0.94	0.67

Our method is $2.7\text{--}4.6\times$ tighter than the product bound ($1.3\text{--}1.5\times$ tighter than plain LipBaB).

Anisotropic per-axis constants then cut the box count by up to **$3.5\times$** in 2D and **$86\times$** in 4D.

Sample-access verification

Two different methods: naive and adaptive.

- Monte Carlo estimation
- Adaptive Multi-Armed Bandit

Issues Even with adaptive sampling, tighter Lipschitz bounds, and parallelisations we were still unable to get consistent terminations within a 10^9 sample budget!

Abstract interpretation (bound propagation)

State-of-the-art methods via Auto-LiRPA.

- **IBP** — cheap, loosest
- **CROWN** — linear relaxation, keeps neuron correlations
- α -**CROWN** — optimised slopes, tightest, costliest

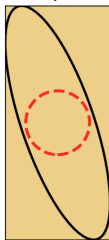
Anytime

Track an upper and lower bound and discretise when the bound straddles zero.

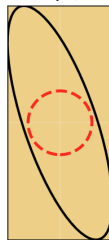
CROWN branch-and-bound refines the domain (Pendulum)

InvertedPendulumLQR($h=0.05$) — crown partition by verification status

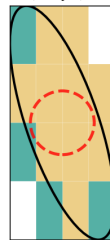
depth 0
(0 verified, 1 open, 0 violating)



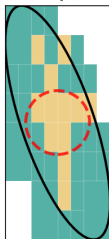
depth 1
(0 verified, 4 open, 0 violating)



depth 2
(4 verified, 10 open, 0 violating)



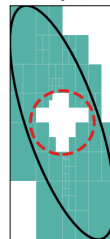
depth 3
(28 verified, 12 open, 0 violating)



depth 4
(54 verified, 6 open, 0 violating)



depth 6
(78 verified, 0 open, 0 violating)



Optimisation perspective

Worst-case drift

$$x^* \in \arg \max_{x \in \mathcal{X}} \mathfrak{D}(x)$$

Verification

$$\mathfrak{D}(x^*) \leq 0$$

a $\forall$ claim — sound, expensive

Refutation

$$\text{any } \mathfrak{D}(x) > 0$$

an $\exists$ witness — a search problem

Refutation as black-box optimisation

Refuting needs only *some* x with $\mathfrak{D}(x) > 0$ — an $\exists$, a search problem.
Soundness not required, so a Monte-Carlo estimate suffices:

$$\tilde{\mathfrak{D}}(x) = \frac{1}{m} \sum_{i=1}^m V(f(x, w_i)) - V(x) + \varepsilon$$

Run as a cheap *pre-screen*; fall back to the verifier only when it finds nothing.

Six optimisers

Uninformed

- **Random** — quality yardstick
- **Grid** — non-adaptive

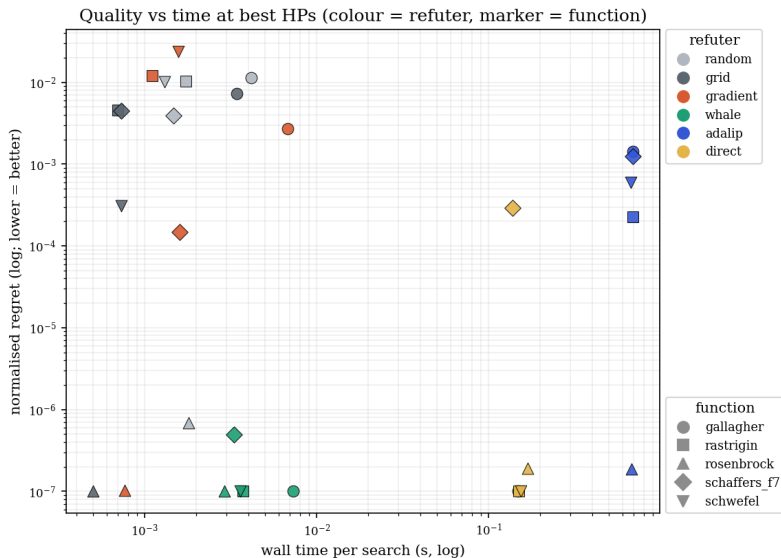
Lipschitz-aware

- **DIRECT** — parameter-free
- **AdaLIPO** — online L

Local / population

- **Hill climbing** — gradient + restarts
- **Whale (WOA)** — population, spiral search

BBOB: quality versus wall-clock cost



Refuting faux certificates

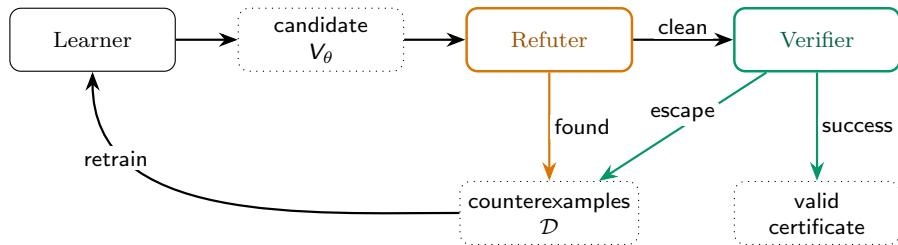
Replay 2169 verifier-rejected candidates against the verifier that rejected each.

Refuter	Found	Speed-up	Quality
Whale	99%	131×	0.049
AdaLIPO	95%	69×	0.042
Random	94%	344×	0.039
Grid	94%	352×	0.038
Hill climbing	91%	262×	0.038
DIRECT	40%	116×	0.024

Median 1–21 ms vs the verifier's 0.48 s.

The refuter pre-screen

The cheap refuter hunts first; the sound verifier runs only on a clean candidate.



Most rounds end at the **refuter**; the **verifier** is paid once, on the final clean candidate.

Results: speed-up tracks the cost of a sound call

Engine	System	Strategy	Time (s)	Rounds	Calls	Speed-up
CROWN	LinSwitch	verifier-only	44.6	11.8	11.8	—
		random	52.2	15.4	2.0	0.85×
		whale	38.5	10.8	1.0	1.16×
α -CROWN	LinSwitch	verifier-only	935.0	69.2	69.2	—
		random	91.8	15.4	2.0	10.19×
		whale	59.6	10.8	1.0	15.70×
	Pendulum	verifier-only	9559.4	8.0	8.0	—
		random	1363.0	6.8	1.0	5.97×
		whale	1331.5	8.8	1.0	6.81×

- Expensive calls $\Rightarrow$ big wins (15.7 $\times$; Pendulum hours $\rightarrow$ minutes)
- Cheap calls $\Rightarrow$ little to save — random can even regress (0.85 $\times$)

Summary & Contributions

- **Verifier tool suite** — a plug-and-play tool for benchmarking CEGIS.
- **Refutation as optimisation** — whale matches verifier quality at 1–3 orders less cost.
- **Central result** — a refuter pre-screen: up to $15.7\times$ faster!
- **Lipschitz bounds** — tighter drift-Lipschitz: $> 80\times$ fewer boxes in 4D linear systems.

Sound verification is the price of a guarantee — but not every round.

Future work

- **Nonlinear drift-Lipschitz** — bound propagation over $V \circ f - V$?
- **High-dimension systems** — can we overcome the curse of dimensionality?
- **Rescue sound sampling** — Tighter bounds, parallelisations
- **Broader specifications** — safety, reach-avoid, etc.
- **Abstractions** — Neural networks are hard to verify by nature

Thank you

Questions?

<https://github.com/will-bailkoski/streamlining-neural-certificates>



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Getting $L_{\mathfrak{D}}$

Lipschitz constants compose

$$L_{f \circ g} \leq L_f L_g \qquad L_{f \pm g} \leq L_f + L_g$$

Composition bound ($\mathfrak{D} = V \circ f - V$)

$$L_{\mathfrak{D}} \leq L_V (L_f + 1)$$

- L_f — **given** (a property of the dynamics)
- L_V — must be estimated from the candidate certificate, *efficiently*

Concept: exact $L_{\mathfrak{D}}$

Assume a linear, deterministic system A ; compose the drift network $\mathfrak{D} = V \circ A - V$ and run LipBaB over its ReLU regions.

Exact drift-Lipschitz constant

$$L_{\mathfrak{D}} = \max \{ \|A^{\top} g_j - g_i\| : R_i \cap A^{-1} R_j \cap \mathcal{X} \neq \emptyset \}$$

- keeps the **cancellation** between $A^{\top} g_j$ and g_i
- maximises over regions i, j **independently**

Anisotropic constants for grid-aligned cells

Per-axis sensitivity, free from LipBaB's per-region gradients:

$$L_V(e_i) = \sup_x \left| \frac{\partial V}{\partial x_i} \right| \leq L_V.$$

On an axis-aligned cell with sides $w_1, \dots, w_d$:

Isotropic

$$|V(x) - V(y)| \leq L_V \|w\|_2$$

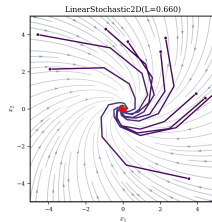
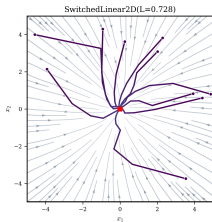
Anisotropic

$$|V(x) - V(y)| \leq \sum_i L_V(e_i) w_i$$

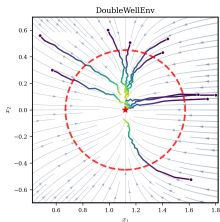
refine per axis $\Rightarrow$ box count $\propto \prod_i L_V(e_i)$ instead of L_V^d

Both contributions together: box count down up to $3.5\times$ (2D) and $86\times$ (4D).

Benchmark phase portraits

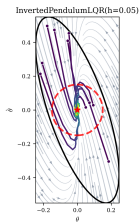


LinSwitch



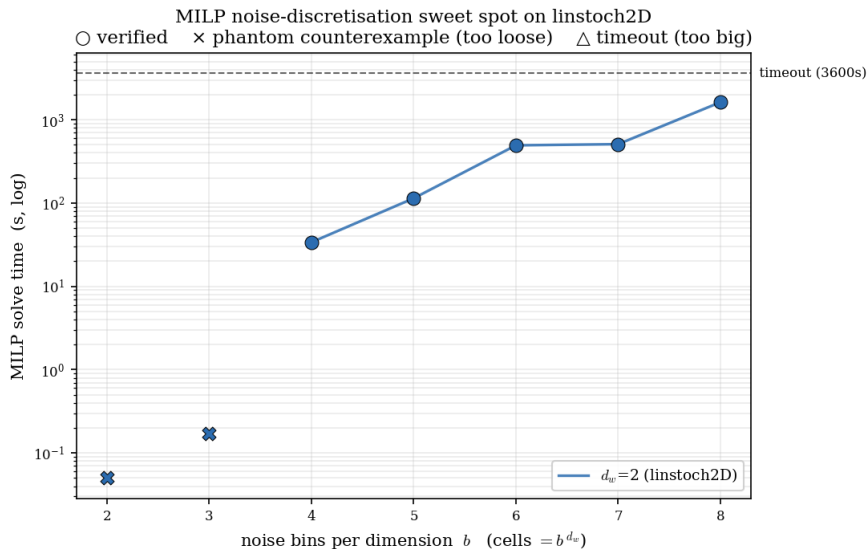
DoubleWell

LinStoch



Pendulum

MILP: the noise-discretisation window



The capability boundary

System	Dynamics	Symbolic	Bound prop.	Sampling
LinSwitch	linear	exact*	✓	sound [†]
LinStoch	linear + noise	exact only ($\leq 2D$)	fails	sound [†]
DoubleWell	polynomial	exact*	✓	sound [†]
Pendulum	transcendental	abstains	only this family	sound [†]

No single family spans the suite — the boundary is *two-sided*.
LiRPA drives the campaign; rejected candidates become the *faux-cert* library.

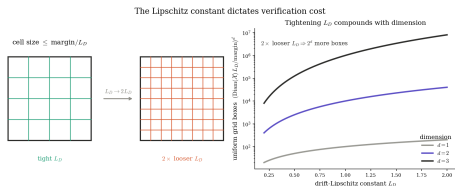
Why it matters for grid verification

Grid the domain; each cell needs
 $L_{\mathcal{D}} \cdot \text{Diam}(B) \lesssim \varepsilon$.

Box count

$$N \approx \left(\frac{L_{\mathcal{D}}}{\varepsilon} \right)^d$$

A d -th power of $L_{\mathcal{D}}$ — tightening compounds with dimension.

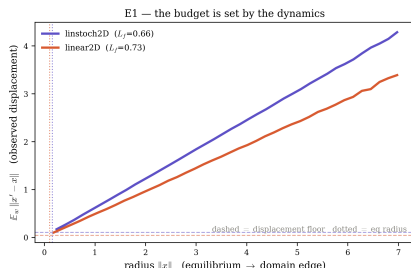


The margin budget

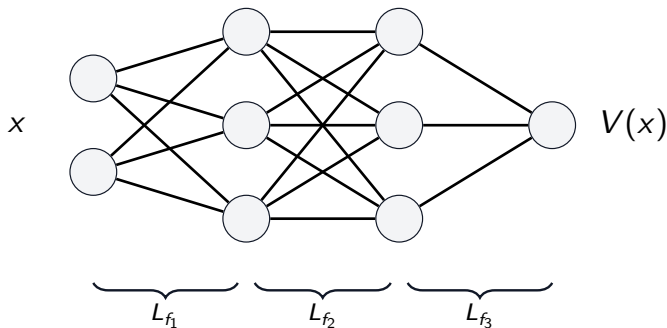
L_V also caps the margin you can certify *at all*:

$$\varepsilon \leq L_V \inf_x \mathbb{E}_{x'}[\|x - x'\|]$$

Danger: too-small L_V (an over-constrained net) makes verification *infeasible* — not just expensive.

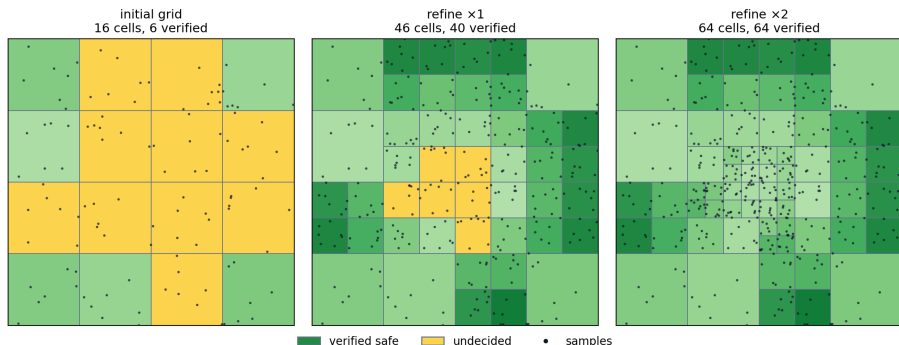


The product-of-layers bound



$$V = f_l \circ \dots \circ f_1 \quad \Rightarrow \quad L_V \leq \prod_{\ell=1}^l L_{f_\ell}$$

The MAB verifier: refine only where it matters



Sample each cell; it **verifies** once its $UCB \leq 0$ (greener = safer), otherwise it *splits*. Refinement concentrates on the thin-margin core.

Full CEGIS results: IBP & CROWN

Engine	System	Strategy	Time (s)	Rounds	Calls	Speed-up
IBP	LinSwitch	verifier-only	155.5	43.8	43.8	—
		random	58.9	18.0	2.6	2.64×
		whale	39.8	10.8	1.0	3.91×
	DoubleWell	verifier-only	14.6	1.3	1.3	—
		random	14.1	1.7	1.0	1.03×
		whale	18.4	2.2	1.0	1.04×
	Pendulum	verifier-only	55.0	6.0	6.0	—
		random	33.7	7.5	1.0	1.76×
		whale	57.4	10.0	1.0	—
CROWN	LinSwitch	verifier-only	44.6	11.8	11.8	—
		random	52.2	15.4	2.0	0.85×
		whale	38.5	10.8	1.0	1.16×
	DoubleWell	verifier-only	41.7	1.7	1.7	—
		random	31.3	2.2	1.0	1.48×
		whale	32.0	2.2	1.0	1.51×
	Pendulum	verifier-only	203.1	7.0	7.0	—
		random	57.8	6.8	1.0	2.74×
		whale	66.8	8.5	1.0	3.19×

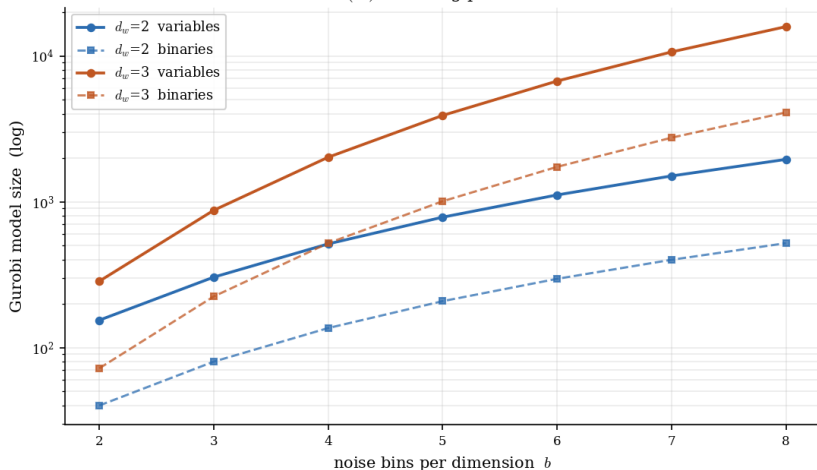
Full CEGIS results: α -CROWN

Engine	System	Strategy	Time (s)	Rounds	Calls	Speed-up
α -CROWN	LinSwitch	verifier-only	935.0	69.2	69.2	—
		random	91.8	15.4	2.0	10.19×
		whale	59.6	10.8	1.0	15.70×
	DoubleWell	<i>did not complete</i>				
	Pendulum	verifier-only	9559.4	8.0	8.0	—
		random	1363.0	6.8	1.0	5.97×
		whale	1331.5	8.8	1.0	6.81×

α -CROWN Pendulum: ~ 20 -min sound calls turn a multi-hour verifier-only run into ~ 22 min.

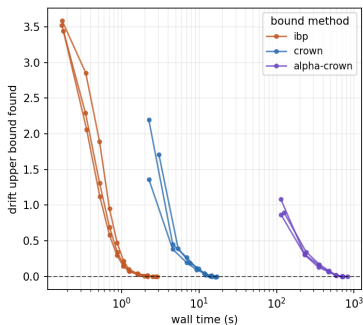
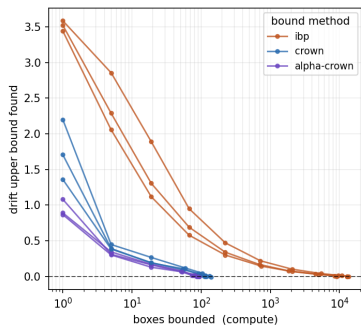
MILP model size grows as b^{d_w}

The MILP grows as b^{d_w} network copies:
one $V(x')$ encoding per noise cell

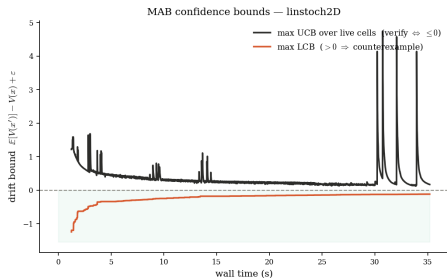
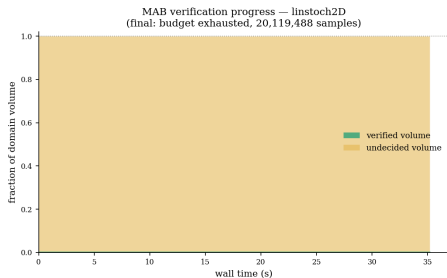


Anytime tightness: IBP / CROWN / α -CROWN

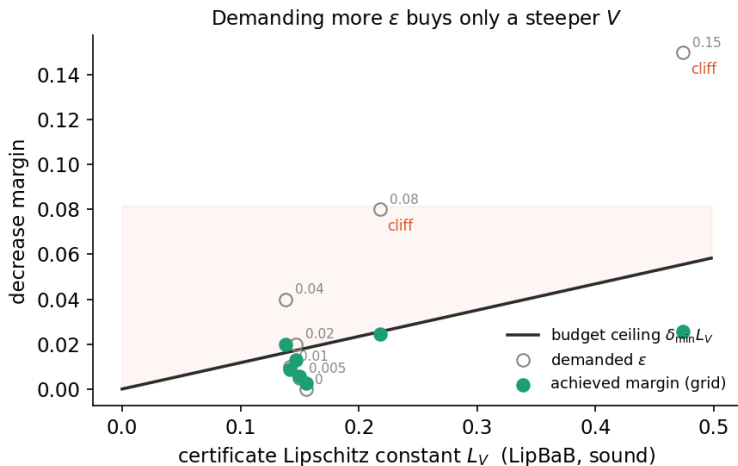
Refuting verified pendulum_lqr certificates: tightness vs compute and vs time (noise_disc=8)



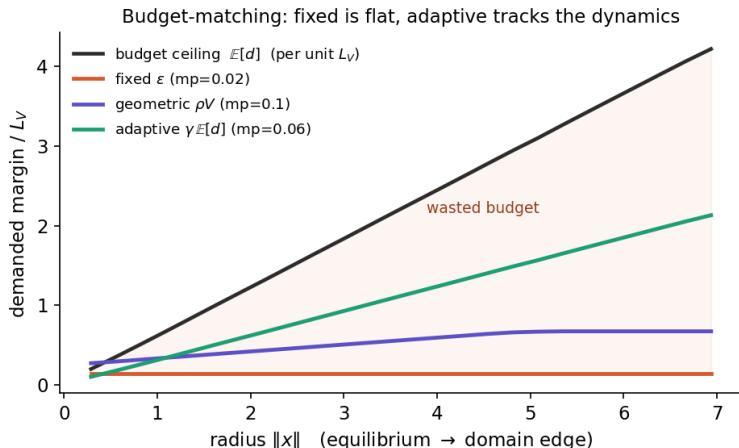
Sound sampling (MAB): coverage and confidence bounds



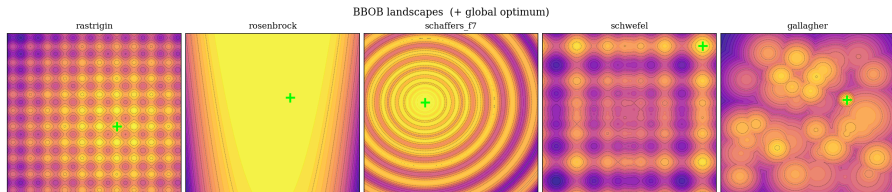
Margin budget: demanding more ε needs a steeper V



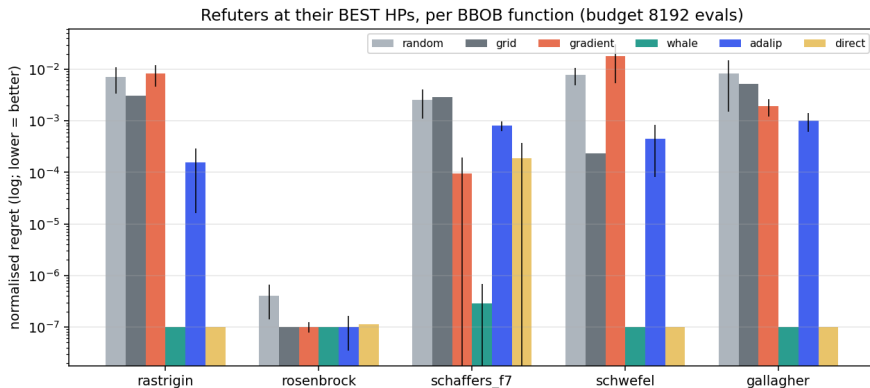
Margin-matching: fixed is flat, adaptive tracks the dynamics



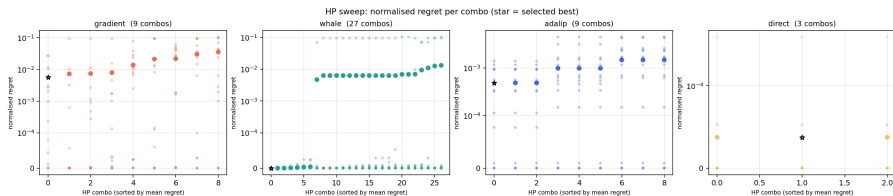
The five BBOB test functions



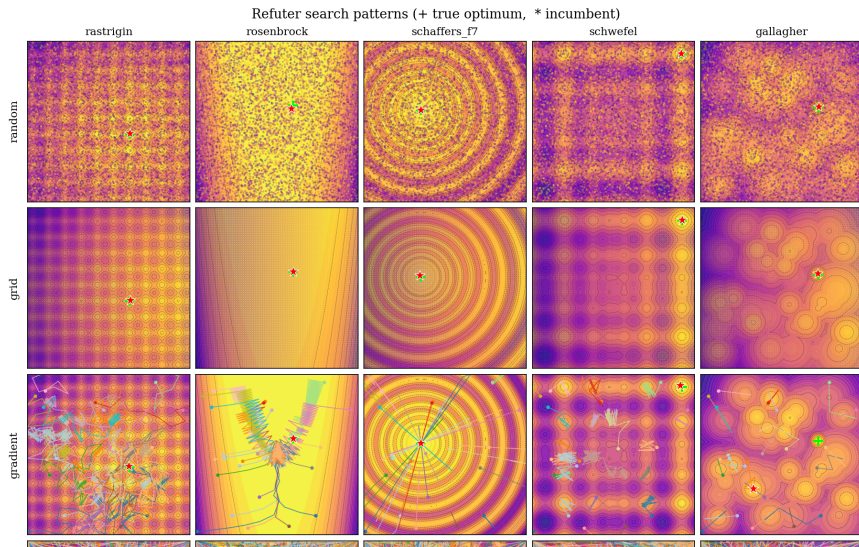
BBOB: normalised regret per function



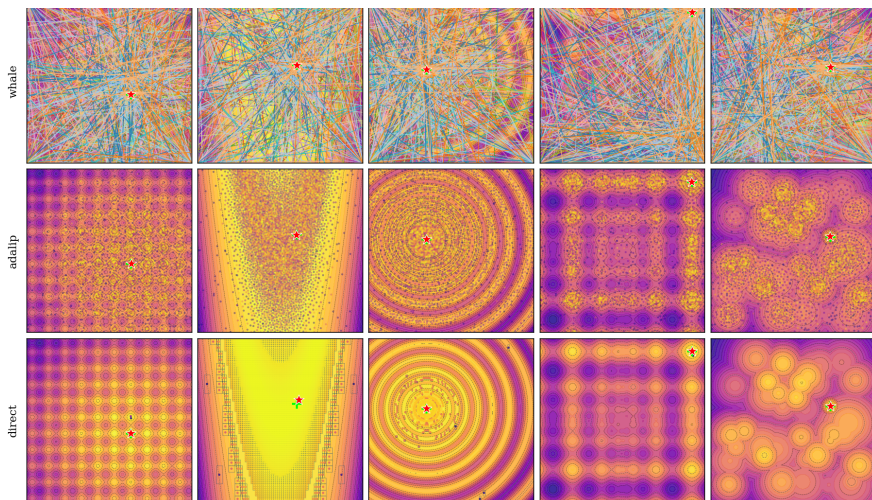
Hyperparameter sweep: regret per combo



Refuter search patterns (I): random, grid, gradient



Refuter search patterns (II): whale, AdaLIPO, DIRECT



columns: Rastrigin, Rosenbrock, Schaffers F7, Schwefel, Gallagher

LinSwitch certificate portrait

